THE INTERNATIONAL UNIVERSITY(IU)
VIETNAM NATIONAL UNIVERSITY - HCMC

FINAL EXAMINATION
PROBABILITY, STATISTICS AND RANDOM PROCESS (MA026IU)
Semester 2, 2024-2025 • June 2025 • Total duration: 90 minutes

Chair of Mathematics Department	Lecturer
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INSTRUCTIONS: *Each student is allowed calculators, statistical tables and ONE double-sided sheet of reference materials (size A4 or similar) marked with their name and ID. All other documents and electronic devices are forbidden.*

Question 1. *(10 pts)* A consignment of 100 half-inch nuts was released from the factory. The diameters of 10 randomly taken nuts were measured. Calculate the sample mean and sample standard deviation using the data in the table.

Number of nut	1	2	3	4	5	6	7	8	9	10
Diameter, inch	0.4983	0.5030	0.4982	0.4986	0.5010	0.5009	0.4984	0.4993	0.5014	0.5010

Question 2. *(15 pts)* The life in hours of a 60-watt light bulb is known to be normally distributed with a standard deviation of 52 hours. In a sample of 80 bulbs, the mean lifetime was 1217 hours. Find a 95% confidence interval for the mean lifetime of this type of light bulb.

Question 3. *(15 pts)* A random sample of 400 electronic components manufactured by a certain process are tested, and 30 are found to be defective. Let p represent the proportion of components manufactured by this process that are defective. Find a 95% confidence interval for p.

Question 4. *(15 pts)* Collect 5-year statistics of the average amount of wheat crop (tons) harvested from $1km^2$ per year, the results are as follows: 560, 525, 496, 543, 499.

Test the hypothesis that the mean wheat crop is 550 tons per year with significant level of 5%.

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Question 5. (*15 pts*) In a random sample of 500 handwritten zip code digits, 464 were read correctly by an optical character recognition (OCR) system operated by the U.S. Postal Service (USPS). USPS would like to know whether the rate is at least 90% correct. Do the data provide evidence that the rate is at least 90% at level of significance 5%.

Question 6. (*30 pts*) Consider a Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{1,2,3\}$ and transition matrix

$$P = \begin{array}{c} \\ \text{From} \\ \end{array} \begin{array}{c} \begin{array}{ccc} \text{To} & & \\ 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0.2 & 0.8 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix} \end{array}.$$

1. Evalute $P(X_4 = 2|X_2 = 1)$ and $P(X_4 = 2, X_2 = 1|X_0 = 3)$.
2. Suppose that $P(X_0 = 1) = P(X_0 = 2) = 0.4$, $P(X_0 = 3) = 0.2$. Compute $P(X_3 = 1)$ and $P(X_5 = 3, X_3 = 1)$.
3. Find the stationary distribution of the Markov chain.